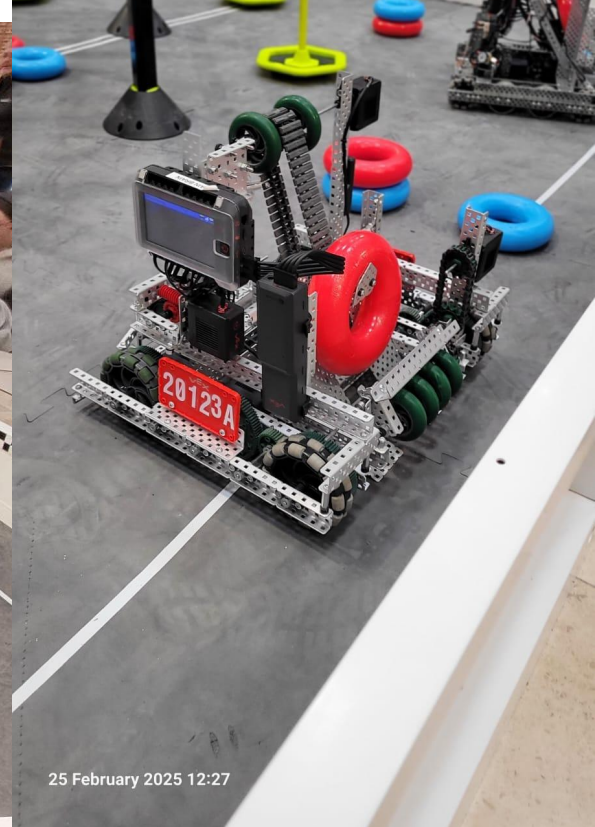
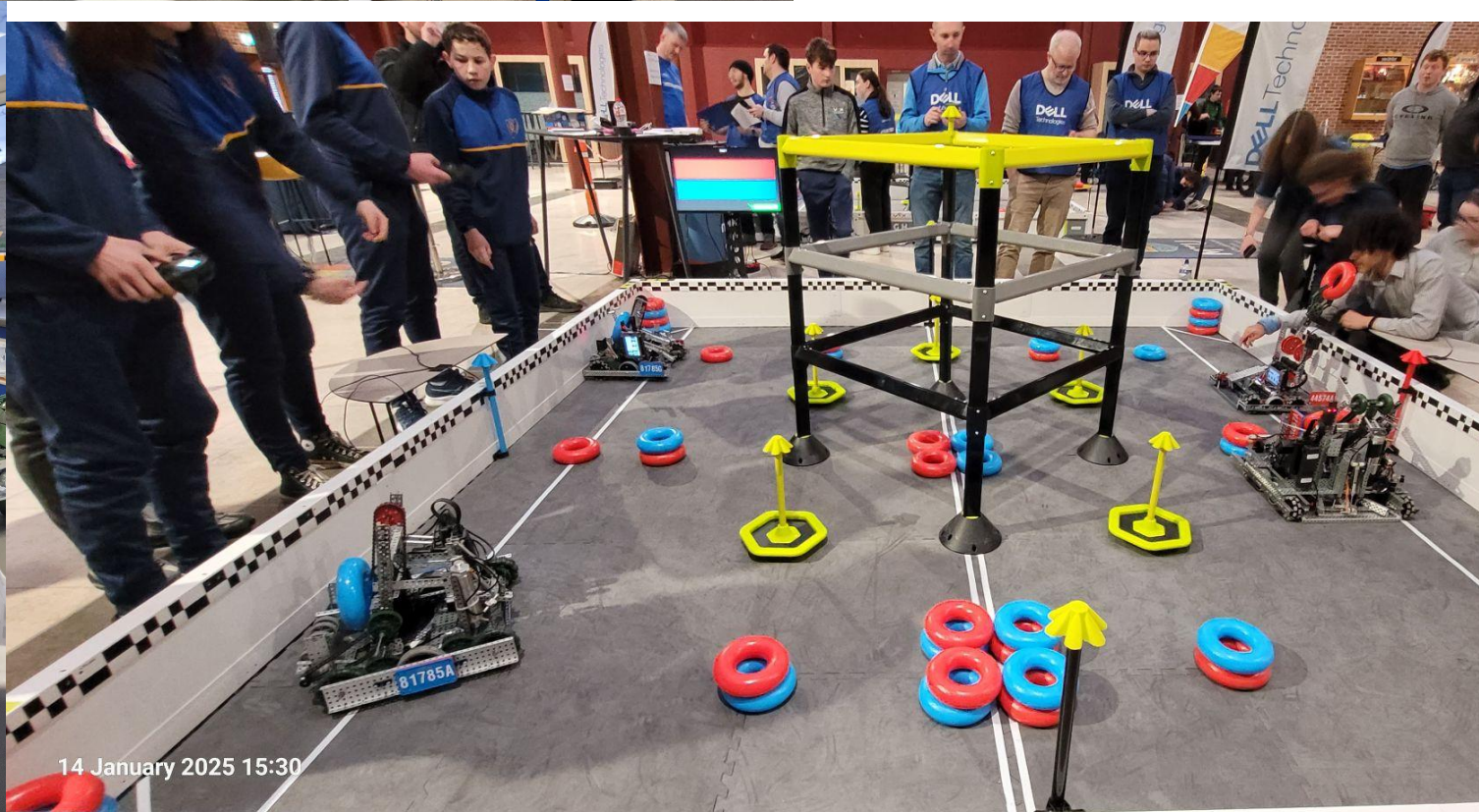
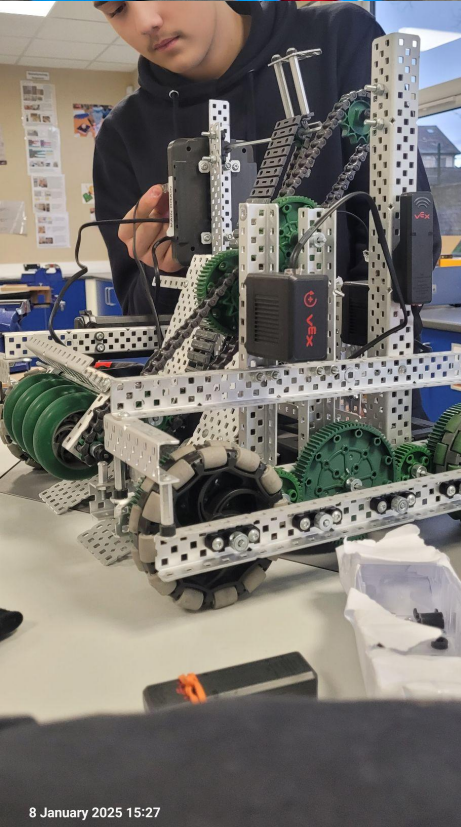
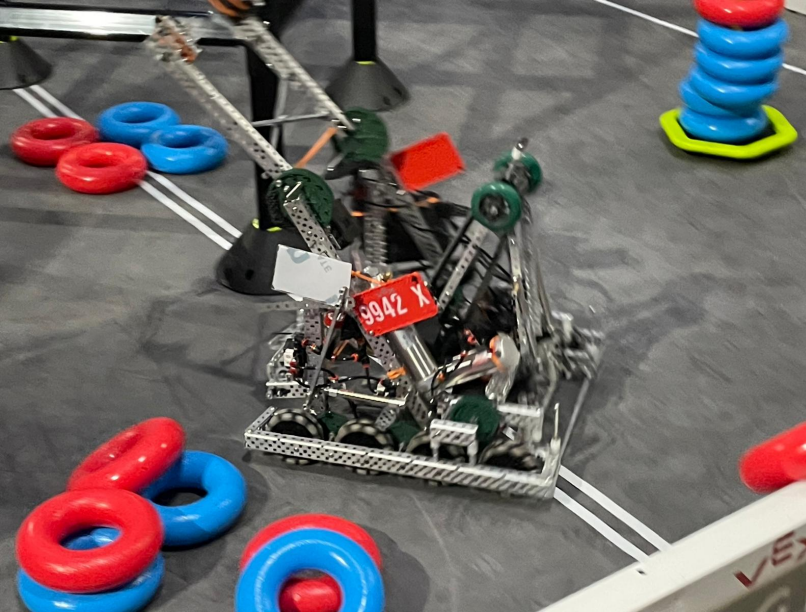


What is VEX Robotics?

VEX and FIRST are intensely competitive arenas where students design, build, and program high-performance robots. It's a high-stakes engineering environment demanding perfect harmony between hardware constraints and software logic.

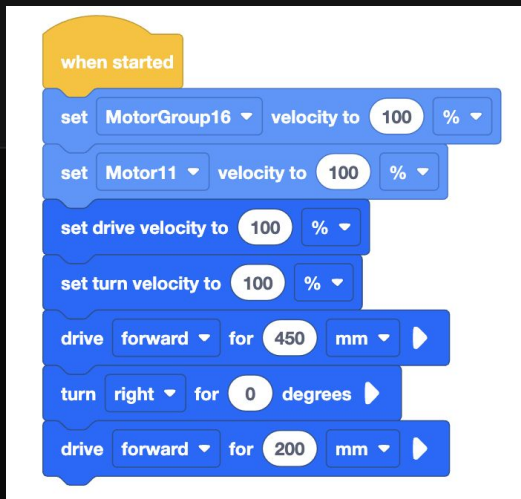


The Coding Friction

Block Code Limits

Most teams start with block coding (e.g., "Move forward 5, turn left").

While accessible, it severely limits potential, making advanced autonomous logic and sensor fusion nearly impossible.



Embedded C++ Hurdles

Transitioning to embedded C++

(PROS/LEMLib) introduces a massive learning curve to find any more potential.

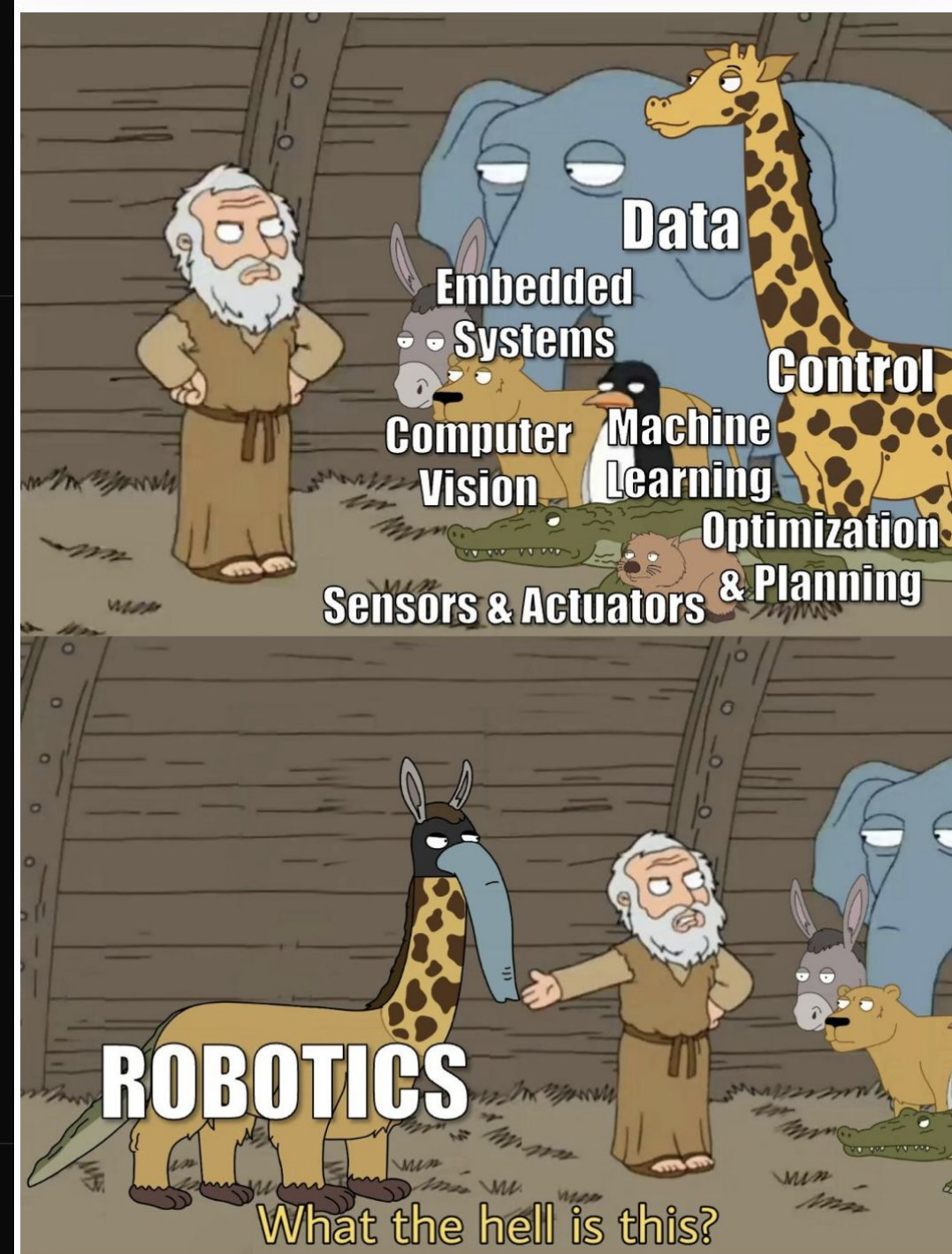
This is a big problem for most teams, leaving them stuck between limited blocks and overly complex code.

The Robotics Reality

Roboticians are like Renaissance engineers. Atoms are harder to control than bits.

- **Software:** Fluency in C++/Python, ROS 2.
- **AI & ML:** Perception, sensor fusion, control.
- **Systems Integration:** Merging sensors, actuators, and embedded systems.
- **Control Theory:** PID, motion planning, trajectory optimization.

Debugging across the hardware/software boundary is an incredibly hard challenge.



Meet Forge



The all-inclusive toolkit for building architecture for VEX and
FIRST bots.

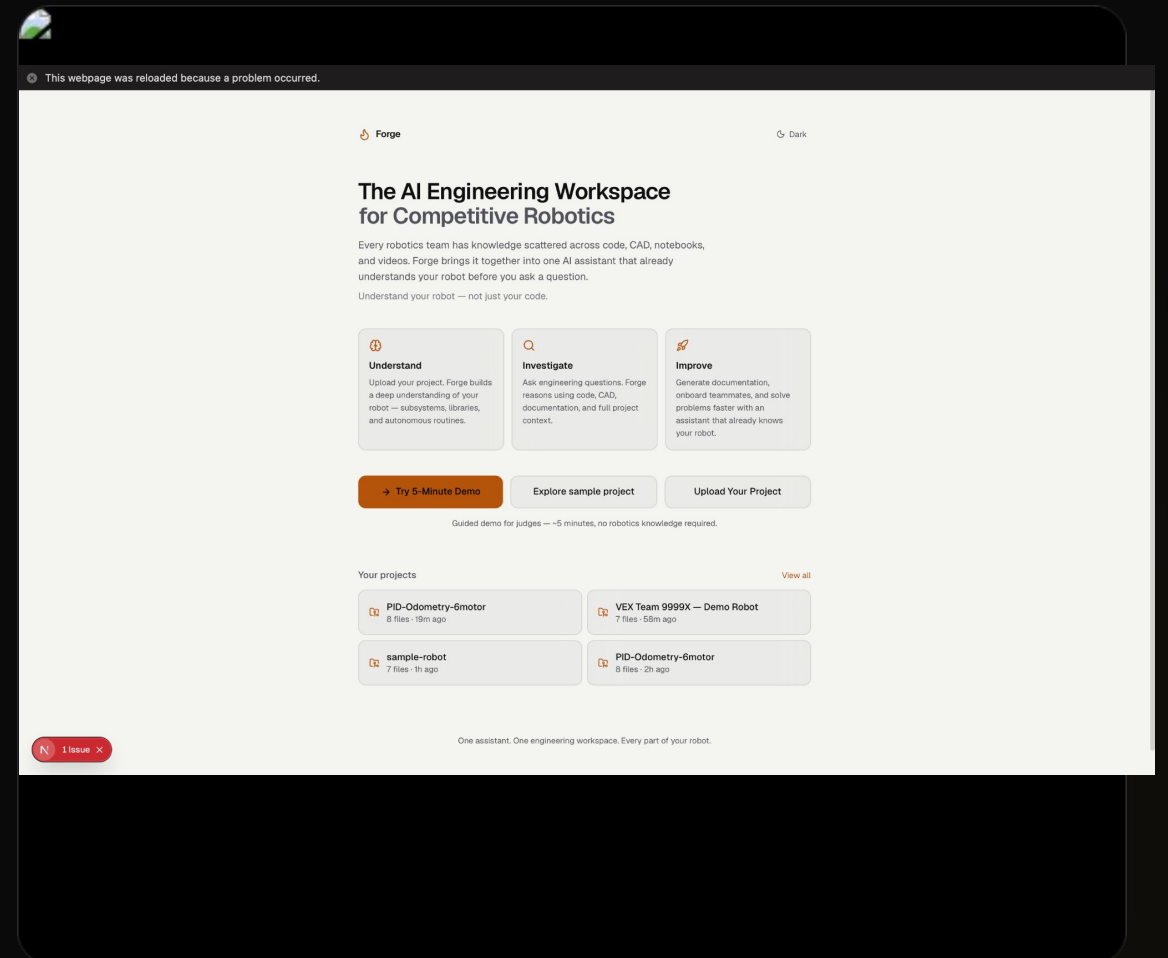
This lets anyone level up their game when it comes to building
competitive bots.

Specialized Context

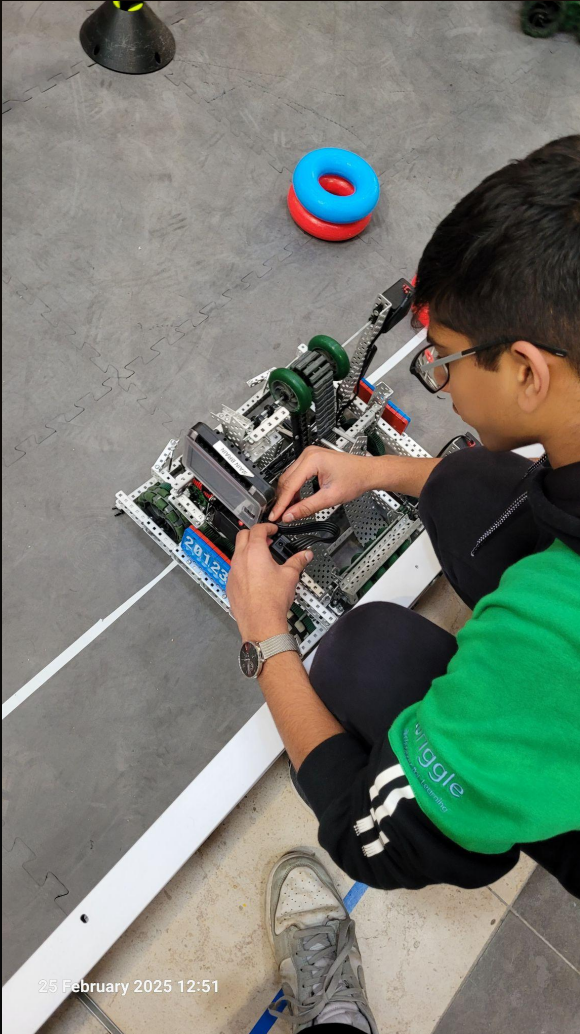
Forge has specialized tools that generic AI like Claude or ChatGPT won't.

It integrates directly with your Onshape CAD, Engineering Notebook, and codebase.

It inherently knows your gear ratios, motor ports, and physical constraints before you ask a single engineering question.



Whisper Flow



25 February 2025 12:51

Analyze Conversations

A primary feature is Whisper Flow, which silently analyzes team conversations and tracks project progress within teams.

Usually, we rely on someone to do it—and mostly it's the guy that forgot to bring food for the team the last meet 🍕🍔.

Let the AI handle the tracking while your team focuses entirely on the build.

The Game Changer

10x

Faster Iteration

Why it's useful

Forge unifies hardware constraints with software logic. Post-match debugging is no longer a game of guesswork.

The AI instantly traces a dropped intake to the exact motor port and anti-jam logic, allowing you to iterate faster than any other team.

How I Built It



Built on advanced context-aware reasoning engines integrating codebase indexing, Onshape API parsing, and real-time telemetry processing to create a true multi-modal engineering workspace.



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